

Complete Java Tutorial – LTS Versions (8, 11, 17, 21, 25)

Java 8 – Functional Programming Foundation

- Lambda Expressions – concise anonymous functions
 - Functional Interfaces – Predicate, Function, Consumer, Supplier
 - Stream API – filter, map, reduce, collect
 - Optional – avoid NullPointerException
 - java.time API – LocalDate, LocalDateTime (immutable & thread-safe)
 - Default & Static methods in interfaces
 - Metaspace replaced PermGen
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Java 11 – Enterprise Stability LTS

- Standard HTTP Client (HTTP/2 + async support)
 - New String APIs – isBlank, strip, repeat, lines
 - Files.readString() and writeString()
 - Flight Recorder (JFR) for production monitoring
 - Removed Java EE & CORBA modules
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Java 17 – Modern Language Improvements

- Records – concise immutable data carriers
 - Sealed Classes – controlled inheritance
 - Pattern Matching for instanceof
 - Switch Expressions
 - Strong encapsulation of JDK internals
 - Security Manager deprecated
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Java 21 – Concurrency Revolution (Project Loom)

- Virtual Threads – lightweight scalable threads
- Record Patterns – destructuring records
- Pattern Matching for Switch (finalized)
- Sequenced Collections – getFirst(), getLast()

- Generational ZGC
 - Structured Concurrency (preview evolution)
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Java 25 – Enterprise Modernization LTS

- Structured Concurrency stabilization
 - Scoped Values (ThreadLocal alternative)
 - Improved GC performance (ZGC/G1)
 - Cloud & container optimizations
 - Security & cryptography enhancements
 - Mature Project Loom ecosystem
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Best Practices & What to Practice

- Use Streams instead of loops where applicable
 - Prefer Records for DTOs
 - Use Sealed Classes for domain modeling
 - Adopt Virtual Threads for high-concurrency applications
 - Upgrade LTS versions progressively (8 → 11 → 17 → 21 → 25)
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